TM-MOE: A Degradation-Aware Transformer-Mamba MoE for Hyperspectral and Multispectral Image Fusion

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***Abstract*—With the increasing necessity for hyperspectral images (HSIs) with fine spatial–spectral details, the hyperspectral (HS) and multispectral (MS) image fusion scheme becomes a consequential mechanism to attain such images due to physical and expense constraints for the standard imaging instruments. With the rapid advancement of deep learning (DL) in the field of image processing, the ability to model long-range dependencies in both HS and MS images has become crucial for achieving superior performance in HS-MS image fusion. Transformer-based models address this challenge effectively through attention mechanisms, whereas state-space models offer a more efficient alternative in terms of both computation and memory usage. Therefore, we devise a degradation-aware transformer-Mamba MoE (TM-MOE) for HS-MS image fusion, which leverages the benefits of the cross-attention mechanism and Mamba to enhance the fusion consequences. The core part of the devised framework is the suggested degradation-aware cross transformer-Mamba (DACTM) model, which leverages the capacity of Mamba in processing long-range relationships with improved training efficiency and the performance of the transformer's feature fusion for the observed pair. Moreover, the suggested Mamba module utilizes a 3-D scan to explore the correlation between pixels in an HSI, and it incorporates a mixture-of-experts (MoE) network to enrich model capability while maintaining effortless active parameter use. While the fusion process is accomplished on various scales, the novel feature accumulation and reconstruction block (FARB) devised reconstructs the final output from the fused features at different scales, gradually. Extensive experiments demonstrate the effectiveness of the proposed TM-MOE. The code of this study will be available on https://github.com/Abdolraheemkhader/TM-MOE**

***Index Terms*—Attention mechanism, hyperspectral image (HSI), image super-resolution, multiscale fusion, spatial–spectral feature scanning.**

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I. Introduction

THE hyperspectral image (HSI) is an image that retains a wealth of spectral knowledge captured over a distinct electromagnetic field with large continuous spectral channels at slim bandwidths [1]. By virtue of its wealth of information, HSIs are increasingly demanded in numerous applications, for example, military pursuits [2], vegetation health monitoring [3], and land cover classification [4]. Unfortunately, owing to their material boundaries, current imaging instruments fail to gather high spatial resolution HSI (HrHSI), which degrades their performance in these applications. Nevertheless, on the bright side, two sorts of images can be captured of the same scene: a high-resolution multispectral image (HrMSI) and a low-resolution HSI (LrHSI). Therefore, merging the complementary details of both HrMSI and LrHSI (known, HS-MS image fusion) evolves the most practical strategy to acquire HrHSI [5], [6], [7], [8]. HS-MS image fusion techniques cope with the physical restrictions and the expenses of attaining HrHSI via the typical sensors. Where both HrMSI and LrHSI possess spatial and spectral attributes, fusing them poses a greater challenge in conserving the substantial details and minimizing spectral and spatial misshaping.

Over the past few years, considerable HS-MS image fusion processes have been introduced, yielding encouraging outcomes. HS-MS image fusion strategies can be organized based on the hypotheses and mechanisms into classical or model-based [9], [10], deep learning (DL)-based [11], and, more recently, transformer-based [12] approaches. The model-based strategies accomplish the fusion task relying on mathematical hypotheses that model the affinities between the observed images and the wanted HrHSI. For example, established on the linear mixture model (LMM), which transforms the fusion procedure into the estimation of two low-rank matrices, dictionary and abundance, that are employed to rebuild the target HrHSI. In this regard, performing the nonnegative matrix factorization (NMF) mechanism, the fusion approaches devised in [10], [13], and [14] calculate the dictionary from the LrHSI and the

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abundance matrix from the HrMSI. Furthermore, by involving various image priors, for example, nonlocal self-similarity and nonnegative sparse coding priors, aiming to enrich further fusion results, further fusion methods have been introduced [15], [16]. Likewise, with accounting for the 3-D property of HSIs, various fusion frameworks are offered that engage the tensor factorization mechanism [9], [17], [18]. Herein, rather than unfolding the inputs into matrices, the feature extraction and fusion are performed in a 3-D manner. However, despite their satisfactory outcomes, model-based practices are hindered by obstacles such as the iterative solution, which makes them more complicated, and handcrafted variables that demand extensive experience.

In recent times, DL has evolved as prevalent in image processing, particularly by utilizing the capacity of the convolutional neural networks (CNNs) in feature extraction. Founded on the CNNs, numerous deep fusion models have been introduced, developed by piling several convolutional layers to pull the vital features from the inputs to reconstruct the target HrHSI [19], [20]. For example, a two-stream network, where each stream piles several convolutional layers that exploit the characteristics of the HrMSI and LrHSI separately (stream for each) and fuse them via fully connected (FC) layers, is devised in [21]. Furthermore, the method developed in [22] fuses the HrMSI and LrHSI through three modules, cross-mode message inserting, spectral and spatial reconstruction units, where the first block performs initial fusion and refines it via spatial and spectral reconstruction blocks. Despite their superior performance, the CNN-based methods have several challenges, for example, their limited kernel size, which fails to exploit the long-range dependencies of the pixels.

In more recent years, the transformer module has gained increased engagement in image analysis, driven by its advantages for modeling nonlocal correlations in large language frameworks that employ the attention mechanism. Therefore, numerous HS-MS image fusion strategies have been proposed based on the transformer, which successfully utilize the intrinsic information, such as nonlocal spatial–spectral correlations (SSCs) and global spectral correlations (GSCs), in image fusion [23], [24]. For example, the concatenated features of the HrMSI and upscaled LrHSI are processed by pixelShuffle and attention blocks to exploit the details of the observed pair to deliver HrHSI in [25]. Le et al. [26] designed MDC-FusFormer, which operates in three units, where the first and second units are for dragging the details, and the third stream is a module based on the attention mechanism for merging the characteristics at various sizes. Herein, the final output is refined by utilizing a multidimensional refinement convolution unit. The transformer-based HS-MS image fusion strategies yield outstanding outcomes, but their memory and computational requirements are more heightened than SSMs (such as Mamba). Therefore, to capture long-range dependence more effectively, state-space models (SSMs), such as the Mamba module, have attained more attention recently [27], [28]. Motivated by the efficient Mamba architecture in terms of training, the Mamba convolutional fusion block that incorporates the attributes extraction capacities of Mamba and CNN for merging the spectral and spatial information dragged by spectral compression and multiscale invertible spatial blocks, is devised in [27]. Although Mamba-based approaches are practical to train with boosted throughput, they still lag behind the transformer's performance [29].

To overcome the aforementioned restrictions, this article proposes a degradation-aware transformer-Mamba MoE for HS–MS image fusion (TM-MOE), which leverages the complementary powers of transformer and Mamba models. TM-MOE progressively rebuilds the desired HrHSI by hierarchically fusing characteristics from the observed images. The core invention lies in the degradation-aware cross transformer-Mamba (DACTM), which integrates 3DMamba blocks with the degradation-aware dual cross-attention (DDuA) transformer module. Precisely, the devised 3DMamba block engages three Mamba layers to scan features in six directions, effectively modeling long-range dependencies, while its final layer integrates a MoE network to balance scalability and efficiency. Meanwhile, the DDuA inspires constrained cross-attention guided by the imaging sensor's spectral responsivity, facilitating the fusion of better, physically consistent information. To complete reconstruction, the feature accumulation and reconstruction block (FARB) employs depthwise separable convolutions and multiscale aggregation to refine both spectral and spatial details. Overall, TM-MOE acquires an adaptable tradeoff between memory efficiency, performance, and adaptability. The contributions of this article are outlined as follows.

1) We propose a TM-MOE, which leverages the capabilities of the transformer and Mamba mechanisms, aiming to strike a balance between the partially conflicting purposes of boosted productivity, minimal memory requirements, and elevated performance in the field of HS-MS image fusion.
2) The DACTM is devised in this article, which incorporates the 3DMamba and DDuA modules. The 3DMamba, which is boosted by incorporating the MoE network, exploits the vital information of the input images employing a 3-D scan (in six directions), while DDuA relies on the cross-attention strategy and degradation for robust feature fusion.
3) We design a simple, effective FARB unit for reconstructing the HrHSI for the outputs at various scales.
4) Extensive experiments are conducted to prove the performance of the devised TM-MOE network, revealing its effectiveness compared to competing testing approaches.

The rest of this article is systematized as follows. Section II inspects the related studies, while the details of the suggested TM-MOE network are demonstrated in Section III. Section IV illustrates the experimental details and examines the outcomes, and finally Section V concludes this article.

II. Related Works

In this part, the three classes (model-driven, data-driven, and transform techniques) for HS-MS image fusion are briefly introduced.

A. Model-Driven Techniques

Model-driven (traditional) HS-MS image fusion strategies primarily depend on the optimization (mathematical) formula

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that describes the affinity between the observed pair and aimed HrHSI [30], [31]. For example, believing the low-rank characteristic of the HSIs, HSIs can be factorized into various low-rank matrices. Therefore, diverse fusion models were devised based on LMM that estimate the HrHSI by multiplying the spectral basis acquired from LrHSI and the coefficients calculated from the HrMSI [32], [33], [34]. The techniques proposed in [10], [16], and [32] employ an NMF strategy for extracting the endmember and the abundance matrix from the input images to rebuild the HrHSI. At the same time, the fusion task is constrained by various image priors, such as nonlocal similarity, as shown in [16]. Moreover, Lin et al. [34] devised a fusion strategy utilizing dual NMF based on a Bayesian model with the absence of PSF. Instead of unfolding the 3-D images into matrices, considerable HS-MS image fusion strategies are invented based on the tensor factorization mechanisms that process the image in 3-D shape [9], [18], [35], [36]. For example, by operating the alternating direction method of multipliers (ADMMs) for tackling the variational framework that calculates the core tensor and corresponding coefficients from the observed pair, the HS-MS image fusion approaches based on tensor factorization are presented in [9] and [36]. While they attained satisfactory outcomes, traditional fusion practices suffer greatly from weaknesses, such as time-consuming handcrafted parameters and predefined imaging systems, for most of them.

B. Data-Driven Techniques

Employing the capability of DL, the researchers invented myriad DL-based image fusion frameworks to handle the weaknesses of the conventional HS-MS image fusion strategies. The DL-based approaches principally rely on stacking diverse layers (e.g., convolutional layers) that learn the mapping function between the wanted HrHSI and the captured images couple. To utilize the spatial–spectral information of the input pair, a 3-D CNN fusion network is offered in [20]. Zhou et al. [19] devised a fusion technique that encodes the observed LrHSI in different hierarchies employing a decoder network. The attained features are utilized alongside the pyramid shape of the HrMSI's information to rebuild the HrHSI gradually. A two-branch network that processes each input individually to exploit the spatial and spectral attributes fully and then merges them through FC layers is introduced in [21]. Moreover, a fully CNN-based approach that possesses three units to generate an initial fusion output, process spatial features, and enhance the spectral information is presented in [22]. One of the primary challenges of CNN-based strategies is their black-box characteristic; therefore, interpretable networks have been invented. The interpretable HS-MS image fusion schemes are developed either by incorporating optimization algorithms and CNNs to work separately [37], [38], [39] or by designing a model that unfolds the iterative solution into a network where each specific layer plays a particular step's role [40], [41], [42]. However, despite their sufficient yields, the CNN-based fusion practices are hindered by the narrow filter dimensions, which ignore vital characteristics and fail to model nonlocal dependencies.

C. Transformer Techniques

Motivated by the triumph of transformer in natural language processing [43] for capturing the nonlocal correlation and long-range dependences, multiple HS-MS image fusion networks have been invented [23], [44]. For example, Jia et al. [45] suggested two paths established on the attention mechanism to extract the global characteristics and utilize the cross-modality concatenation unit to merge the attained information. The MIMO-SST that employs multihead information channels and map attention units, gated network, and reconstructor block to fuse the inputs in various sizes and deliver merged images with diverse scales is offered in [24]. Moreover, based on a U-shaped architecture, the HMF-Former is designed, which incorporates a spatio-spectral transformer unit for merging crucial characteristics of the observed pair [46]. Furthermore, an MIMFormer framework, which relies on a model that integrates CNN and the attention mechanism, is presented in [47]. Recent research uses multiscale transformer designs to better leverage spectral–spatial complementarity. To enhance cross-scale dependency modeling, SiT [48], for instance, uses a scale-interaction encoder that extends conventional self-attention with additional scale tokens and extracts spatial and spectral information at different scales. To further ensure balanced spectral and spatial learning, a specific consistency loss is added. The transformer-based practices perform well, while the computation is more complicated compared to recent SSMs, for example, the Mamba model. Therefore, numerous fusion networks have been suggested that borrow the Mamba's efficiency. For example, He et al. [28] developed a pan-sharpening approach inspired by the capacity of the Mamba module in nonlocal characteristics extraction. Moreover, an MCFM which incorporates a CNN and Mamba's attributes acquisition ability is demonstrated in [27]. To lead the enrichment process of characteristics, this unit is developed to improve the complementarity between nonlocal and local characteristics in an adaptive manner. The Mamba networks reveal efficiency and increasing throughput; yet, in performance, it is inferior to the transformer approaches. Therefore, integrating the transformer and Mamba modules to mitigate their limitations and aid each other in gaining adequate yields is a potential area for improvement.

III. Proposed Method

A. Problem Formulation

Let us denote the desired high spatial–spectral resolution hyperspectral image (HrHSI) by $\mathbf{X} \in \mathbb{R}^{L\times M\times N}$, where M and N are the number of pixels in height and width, respectively, and L is the number of spectral bands. Although the HrHSI contains abundant information that makes it highly required in many applications, such an image is not attainable due to physical restrictions. Fortunately, two types of available sensors can capture two dissimilar kinds of images for the identical scene. One of the images has a high spectral resolution with a low spatial resolution (LrHSI), which can be represented by $\mathbf{Y} \in \mathbb{R}^{L\times m\times n}$ ($m = M/a$ and $n = N/a$). Herein, a represents the upscaling factor. The second one has a high spatial resolution with a much lower spectral resolution compared to LrHSI

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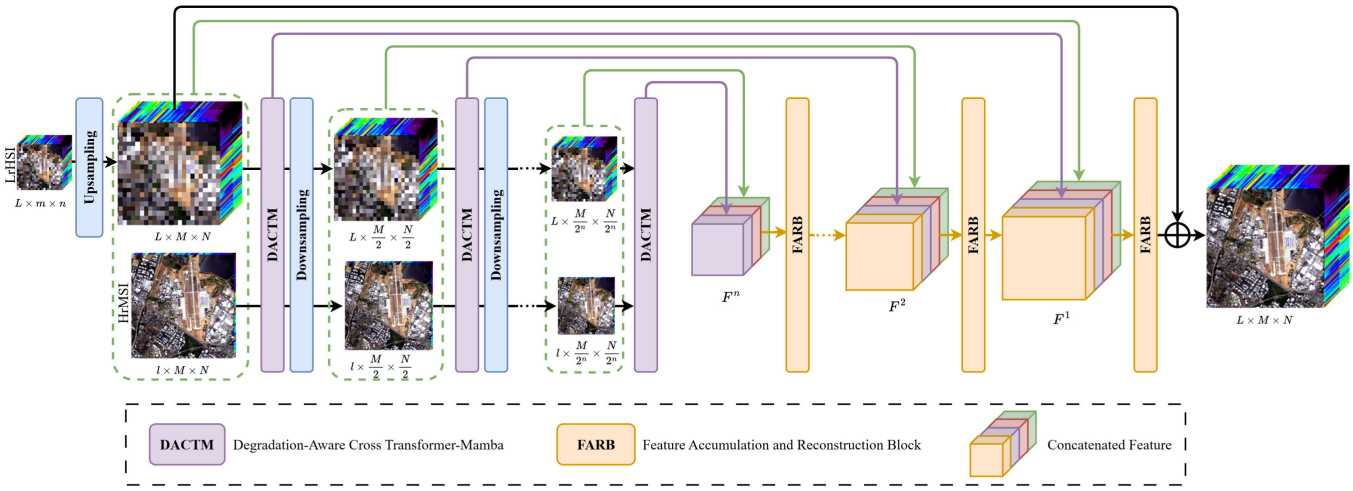


Fig. 1. Overall architecture of the suggested TM-MOE. F is the concatenated features acquired via the DACTM block. This network generates the features from the inputs at three scales and develops corresponding fused features employing the constrained cross-attention mechanism. Here, upsampling is performed using bicubic interpolation, while downsampling is performed using an average pooling layer with a kernel size of 2. The desired HrHSI is reconstructed gradually via the reconstruction block, as illustrated on the left side of the figure.

(HrMSI), which $\mathbf{Z} \in \mathbb{R}^{l \times M \times N}$ can denote ($l < L$ is the number of bands). HrMSI and LrHSI have complementary information that can be utilized to reconstruct the desired HrHSI through image fusion techniques. Herein, the LrHSI is considered a spatial downsampled version of HrHSI, which is expressed as

$$\mathbf{Y} = \mathbf{XB} \tag{1}$$

where $\mathbf{B} \in \mathbb{R}^{MN \times mn}$ represents the degradation blur matrix, which is reconstructed by convolving the spectral bands of HrHSI and the imaging instrument's point spread function (PSF), followed by a downgraded operator. In the same way, the HrMSI can be viewed as a spectrally degraded version of HrHSI, which can be obtained as

$$\mathbf{Z} = \mathbf{RX} \tag{2}$$

where $\mathbf{R} \in \mathbb{R}^{l \times L}$ denotes the spectral downgraded matrix, which is commonly simulated using the spectral response function (SRF) of the corresponding multispectral imaging sensor. Over the past years, many HS-MS image fusion techniques have been proposed based on mathematical assumptions, which are solved analytically or iteratively. However, the performance of these approaches is hindered by many constraints, for example, computation complexity, hand-crafted parameters, and the unavailability of PSF and SRF. With the success of DL in various domains, including but not limited to computer vision, different learning-based HS-MS image fusion models have been developed employing the powerful ability of CNNs, transformers, or recently selective state-space layers (Mamba). The well-trained DL model receives two inputs (LrHSI and HrMSI) to reconstruct the desired HrHSI, which can be represented as follows:

$$\widetilde{\mathbf{X}} = f_{\Theta}(\mathbf{Y}, \mathbf{Z}) \tag{3}$$

where $f_{\Theta}(\cdot)$ is the deep network (based on CNNs, transformer, or Mamba) with Θ learnable parameters. $\widetilde{\mathbf{X}}$ denotes the output of a well-trained $f_{\Theta}(\cdot)$. Although DL technologies cope effectively with mathematical-based strategies' limitations, each has its own restrictions. Transformer models overcome the shortcomings of CNNs regarding nonlocal features, while their quadratic computation complexity and high memory requirements affect their performance and throughput. However, the efficiency of Mamba with linear computation complexity is nonetheless lagging behind the undertaking of comparably sized transformers. Inspired by the advantages of transformer and Mamba, TM-MOE: A Degradation-Aware Transformer-Mamba MoE for Hyperspectral and Multispectral Image Fusion is proposed to mitigate their restrictions and improve the suggested HS-MS image fusion model's performance. The following subsections provide detailed details of the technique offered.

B. Overall Model Architecture

The overall structure of the proposed degradation-aware transformer-Mamba MoE (TM-MOE) is depicted in Fig. 1, where the fusion is performed at multiscale resolution, and the final fused image is reconstructed gradually. The central part of the devised TM-MOE is the novel offered DACTM block that exploits the details of the inputs in different scales through the 3DMamba modules and fuses them via the devised dual cross-attention fusion mechanism. The proposed DACTM comprises three 3DMamba modules and one DDuA transformer block, where the two first 3DMamba are employing to exploits the features of the inputs, and then applying the offered degradation-aware dual cross-attention (DDuA) transformer for merging the extracted information that effectively will refining through the third 3DMamba block (please see Fig. 2). Starting from the smallest scale, the fused features of the encoder are combined with inputs and go through the reconstruction block to gradually build the desired fusion image. The details of the suggested blocks are provided in Sections III-C and III-D.

C. Degradation-Aware Cross Transformer-Mamba

The core part of this proposed network is the developed DACTM block, whose architecture is displayed in Fig. 2.

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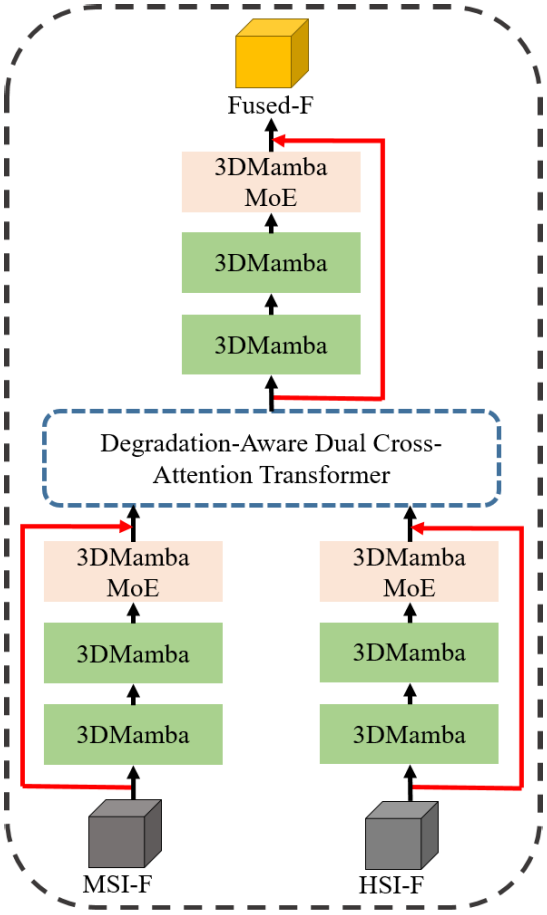


Fig. 2. Illustration of the proposed DACTM. The proposed DACTM module extracts features from the inputs employing three 3DMamba layers (two with FFN and one layer with MoE network). Then, the proposed DDuA transformer network is applied to generate fused features, which are passed through another 3DMamba block to reconstruct the output at this stage.

The DACTM block incorporates two essential parts: the 3DMamba block and the DDuA transformer, which leverage the advantages of both Mamba and the transformer and relieve their adverse side effects. Although the tremendous vogue of the transformer architecture in computer vision and image processing tasks is inspired by its performance in language models, it has two essential disadvantages. First, its high memory consumption limits the ability to handle long contexts. Second, the absence of a single abstract leads to inferior throughput and inference, where every produced token computes the whole context. On the other hand, SSM models (e.g., Mamba) can process long-range relationships with better training efficiency. However, compared with transformer, it still drags rearward in terms of performance. Therefore, the DACTM module benefits from both transformer and SSM clans, as shown in Fig. 2. Due to the sophistication of this part, we discuss it in the following subsections: 3DMamba block and DDuA transformer.

1) 3DMamba Block: As shown in Fig. 2, the proposed DACTM module retains three identical blocks based on the Mamba layer. Each block comprises three mamba layers; the first two include a feedforward network (FFN), while the third layer incorporates a mixture of experts (MoE). Two Mamba blocks serve as feature extractors from the observed

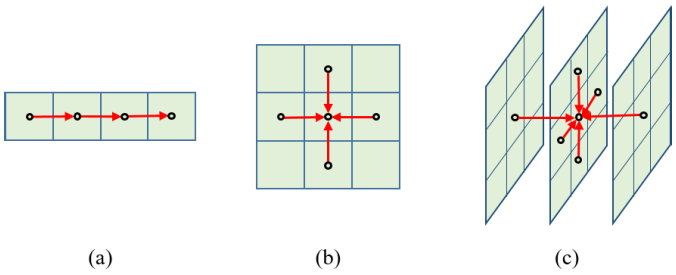


Fig. 3. Demonstration of the dependencies in (a) vanilla Mamba, (b) spatial dependencies in HSIs, and (c) spatial–spectral dependencies in HSIs.

pair before employing the proposed DDuA transformer. At the same time, the third one is utilized to strengthen the fused features generated via the DDuA transformer. This proposed architecture endeavors to reduce the number of active parameters and occupy considerably less cache memory for processing **KV** than the vanilla transformer. Given the input sequence $x(t)$, SSMs map it to a latent expression $h(t)$ and then calculate the outcome sequence $y(t)$. This can be depicted mathematically as follows:

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{4}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$, $\mathbf{B} \in \mathbb{R}^{N\times 1}$, and $\mathbf{C} \in \mathbb{R}^{1\times N}$ are trainable parameters. However, the discrete version of SSMs, which is favorable for DL frameworks, can be acquired by employing the zero-order hold (ZOH) rule and a timescale parameter $\Delta \in \mathbb{R}$. The discrete version can be described as the following:

$$\begin{aligned} h_t &= \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t \\ y_t &= \mathbf{C}h_t \end{aligned} \tag{5}$$

herein, $\bar{\mathbf{A}}$, $\bar{\mathbf{B}}$, and $\mathbf{C}$ represent the discrete variables. The relationship between $\{\mathbf{A}, \mathbf{B}\}$, and $\{\bar{\mathbf{A}}, \bar{\mathbf{B}}\}$ can be expressed as

$$\begin{aligned} \bar{\mathbf{A}} &= e^{\Delta\mathbf{A}} \\ \bar{\mathbf{B}} &= (\Delta\mathbf{A})^{-1}\left(e^{\Delta\mathbf{A}} - \mathbf{I}\right) \cdot \Delta\mathbf{B} \end{aligned} \tag{6}$$

where $\mathbf{I}$ is an identity matrix. This operation can be transferred into a global convolution process to accelerate the calculation, which can be depicted as follows:

$$y = x * \bar{\mathbf{K}} \tag{7}$$

where $\bar{\mathbf{K}} = (\mathbf{C}\bar{\mathbf{B}}, \mathbf{C}\bar{\mathbf{A}}\bar{\mathbf{B}}, \mathbf{C}\bar{\mathbf{A}}^2\bar{\mathbf{B}}, \ldots, \mathbf{C}\bar{\mathbf{A}}^{N-1}\bar{\mathbf{B}}) \in \mathbb{R}^L$ denotes the structured convolutional filter, and * represents the convolutional process. According to (4), all parameters throughout the sequence are assumed constant, discouraging the model from capturing the dynamic relevancy of information at various time steps. Nevertheless, adapting to context-specific features in real-world scenarios that evolve over time is vital. Mamba overwhelms this limitation by utilizing a selective scanning strategy, where the parameters $\mathbf{B}$, $\mathbf{C}$, and Δ become input-dependent parameters, letting the model tailor its behavior to the content of the sequence.

Although the Mamba module performs well in modeling the sequence dependencies, its unidirectional scan method can degrade performance when applied to HSIs to capture the nonlocal spatial correlation, where each pixel correlates with

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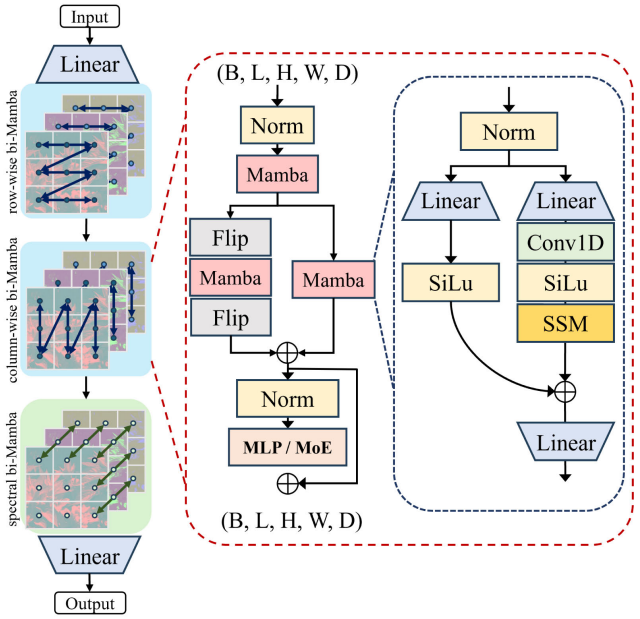


Fig. 4. Illustration of the devised 3DMamba block. The red box on the right side of the figure contains the bidirectional scanning strategy.

its neighbors as depicted in Fig. 3. Therefore, a bidirectional scanning layer is devised for each 3DMamba layer at the Mamba blocks presented in the proposed DACTM module. The offered 3DMamba layer provides an effective and efficient scanning approach based on the Mamba's capability for modeling sequences and mitigates multidirection limitations at the one pixel when using vanilla Mamba. Applying the proposed 3DMamba block ensures the processing pixel considers its neighbors in six directions. The proposed 3DMamba block can be mathematically described as

$$\begin{aligned} E_1 &= \text{Linear}\,(E) \\ E_{rw} &= BiMamba\,(rearrange_{rw}\,(E_1)) \\ E_{cw} &= BiMamba\,(rearrange_{cw}\,(E_{rw})) \\ E_{sw} &= BiMamba\,(rearrange_{sw}\,(E_{cw})) \\ \tilde{E} &= \text{Linear}\,(E_{sw}) \end{aligned} \tag{8}$$

where E and $\tilde{E}$ represent the input and output features, respectively, and *BiMamba* is the devised bidirectional scanning layer as depicted on the right side of Fig. 4. *rw*, *cw*, and *sw* denote row-wise, column-wise, and spectral-wise, respectively. Accordingly, the rows are scanned forward and backward, the columns are scanned in both directions, and the two directions across the spectral dimension as well, as demonstrated in Fig. 4. Formally, given any feature at specific direction (G), the basic Mamba layer can be expressed as

$$\begin{aligned} G' &= \text{Norm}\,(G) \\ G_1 &= \text{SSM}\big(\text{SiLu}\big(\text{Conv1D}\,\big(\text{Linear}\,\big(G'\big)\big)\big)\big) \\ G_2 &= \text{SiLu}\big(\text{Linear}\,\big(G'\big)\big) \\ G_{out} &= \text{Linear}\,(G_1 + G_2)\,. \end{aligned} \tag{9}$$

As shown on the right side of Fig. 4, the outcome of the basic Mamba layer (G_{out}) is fed through two Mamba layers in opposite directions. This yields two features for scanning

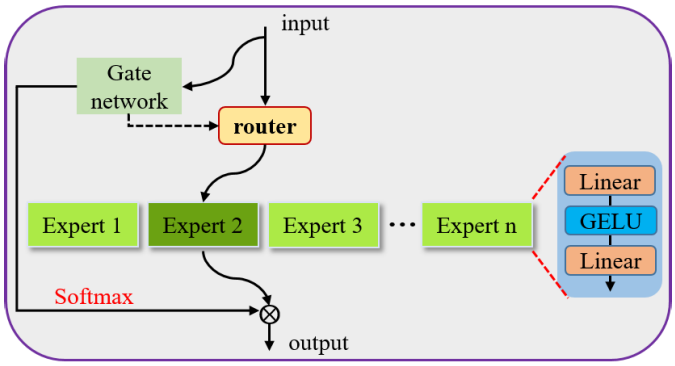


Fig. 5. Demonstration of an MoE network. For any input feature, the router picks one expert to accomplish estimations. The selection depends on the gating network's outcome (speckled line). The final output of the MoE is produced by the multiplication of the softmax of the results of the gating network and the outcome of the employed expert layer (dark-green box).

in forward and backward movements (the same process is employed for rows, columns, and the spectral dimension, resulting in six directions) that form our bidirectional scanning approach. After that, the two acquired features from the opposite directions are then aggregated. These processes can be depicted as below

$$\begin{aligned} G_{fwd} &= \Phi\,(G_{out}) \\ G_{bwd} &= \Phi\,(G_{out}.flip)\,.flip \\ G_{agg} &= \big(G_{fwd} + G_{bwd}\big)\,/2 \end{aligned} \tag{10}$$

where Φ denotes Mamba layer with same operations as in (9). G_{fwd} and G_{bwd} are the scanned features in forward and backward directions for the specific order (e.g., the forward scan traverses the feature rows in raster-scan order from the first to the last pixel, while the backward scan flips the feature map along the row (height) dimension. The same scanning strategy, including forward and backward passes via flipping along the column (width) and spectral dimensions, is applied to ensure bidirectional scanning across all three dimensions.), respectively.

Given the aggregated scanned features, G_{agg}, a normalization layer is applied, and then employing an MLP layer to refine the final output's local texture as follows:

$$\begin{aligned} G' &= \text{Norm}\,\big(G_{agg}\big) \\ G_{mamba} &= \text{MLP}\,\big(G'\big) + G_{agg} \end{aligned} \tag{11}$$

where G_{mamba} denotes the output of the bidirectional scanning layer. In particular, the six-directional scanning is acquired via a bidirectional row scan of the input features. Then, the output of the row scan is processed through a bidirectional column scan, and finally, the results are fed to a bidirectional spectral scan, as described on the left side of Fig. 4. As shown in Fig. 2, each Mamba block comprises three identical 3DMamba (where each has three bidirectional scanning layers) employing an MLP network, except the last layer for each block incorporates a mixture of experts (MoE) rather than an MLP network. The MoE layer expands the model's capability while keeping a favorable number of active parameters and not raising computation complexity. The details of MoE are

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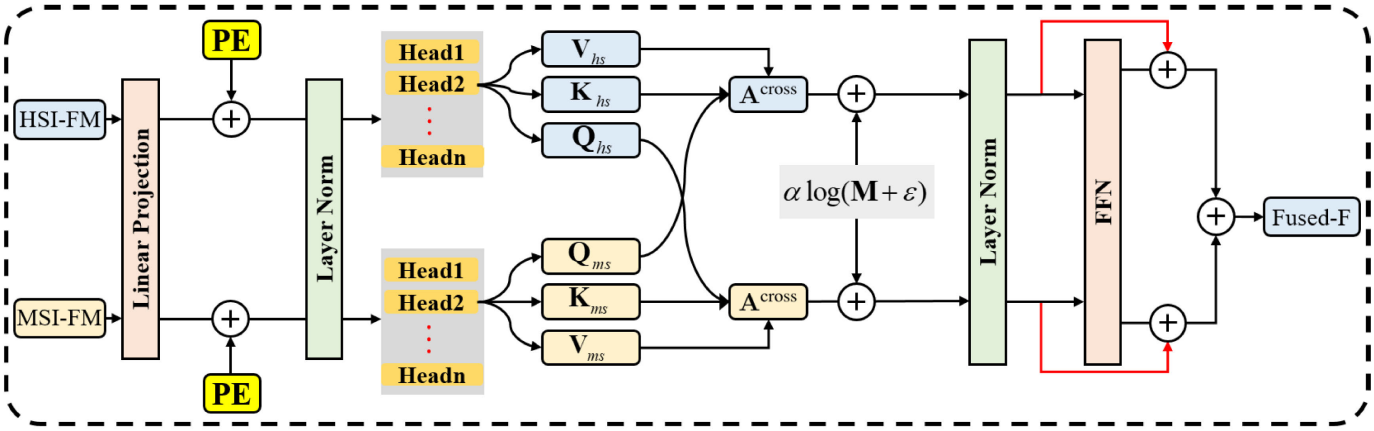


Fig. 6. Architecture of the proposed DDuA transformer block. **M** is a spectral affinity prior, which is estimated by normalizing the SRF matrix of the imaging device. At the same time, α and ε denote the balancing parameter and the constant controlling log instability, respectively.

depicted in Fig. 5, and we set the number of experts in this study to 4.

2) DDuA Transformer: As depicted in Fig. 6, the two extracted features of the two Mamba blocks (denoted HSI-FM ($\mathbf{Z}_1$) and MSI-FM ($\mathbf{Z}_2$)) pass through linear and normalization layers to generate the queries (**Q**), values (**V**), and keys (**K**). Formally, this process can be presented as follows:

$$\begin{aligned} \mathbf{Q}_{hs} &= \mathbf{W}_{\mathrm{Qhs}}\mathbf{Z}_1, \quad & \mathbf{Q}_{ms} &= \mathbf{W}_{\mathrm{Qms}}\mathbf{Z}_2 \\ \mathbf{K}_{hs} &= \mathbf{W}_{\mathrm{Khs}}\mathbf{Z}_1, \quad & \mathbf{K}_{ms} &= \mathbf{W}_{\mathrm{Kms}}\mathbf{Z}_2 \\ \mathbf{V}_{hs} &= \mathbf{W}_{\mathrm{Vhs}}\mathbf{Z}_1, \quad & \mathbf{V}_{ms} &= \mathbf{W}_{\mathrm{Vms}}\mathbf{Z}_2 \end{aligned} \tag{12}$$

where $\mathbf{W}_{\mathrm{Qhs}}, \mathbf{W}_{\mathrm{Qms}}, \mathbf{W}_{\mathrm{Khs}}, \mathbf{W}_{\mathrm{Kms}}, \mathbf{W}_{\mathrm{Vhs}}$, and $\mathbf{W}_{\mathrm{Vms}}$ denote the trainable projection weights. Afterward, we utilize the constrained cross-attention to generate two fused feature versions. Herein, in order to improve the physical interpretability and spectral consistency of characteristic fusion in the HS-MS image fusion problem, we introduce a constrained cross-modality attention mechanism that incorporates domain-specific spectral priors into the attention calculation. Conventional cross-attention algorithms often neglect the fundamental spectrum correlations controlled by the sensor's SRF in favor of calculating dependencies only based on feature similarity. On the other hand, the offered strategy captures the physical overlap and correlation between LrHSI and HrMSI's spectral channels by introducing a spectral affinity prior (**M**), which is derived from the SRF matrix. Particularly, given the SRF ($\mathbf{R} \in \mathbb{R}^{l\times L}$), where each element represents the contribution of a specific LrHSI's spectral channel to the specific HrMSI's spectral band, the spectral affinity matrix is calculated by normalizing **R** as follows:

$$\widehat{\mathbf{M}} = \frac{\mathbf{R}}{\sum_{i=1}^{l}\sum_{j=1}^{L}\mathbf{R}_{i,j}}. \tag{13}$$

In order to integrate the spectral prior into the spatial cross-attention framework without adding to the computational overhead, a scalar degradation-aware prior is derived from $\widehat{\mathbf{M}}$, which mirrors the aggregate spectral compatibility between the multispectral and hyperspectral detectors, as

$$\mathbf{M} = \frac{1}{lL}\sum_{i=1}^{l}\sum_{j=1}^{L}\widehat{\mathbf{M}}_{i,j}. \tag{14}$$

After that, **M** is replicated to fit $\mathbf{Q}_{ms}\mathbf{K}_{hs}^{\mathrm{T}}$'s dimension, and the formal calculation of the attention weights is as follows:

$$\mathbf{A}_{hs}^{\mathrm{cross}} = \mathrm{softmax}\left(\frac{\mathbf{Q}_{ms}\mathbf{K}_{hs}^{\mathrm{T}}}{\sqrt{d}} + \alpha\log\left(\mathbf{M}+\varepsilon\right)\right)\mathbf{V}_{hs} \tag{15}$$

where α and ε represent the balancing parameter and the constant controlling log instability, respectively. This formulation operates as a degradation-aware regularization term, where the attention distribution is globally modulated by the logarithmic affinity bias in harmony with the spectral compatibility enforced by the SRF. The injected spectral bias boosts the attention mechanism to prioritize interchanges associated with spectrally faithful channels, thereby sustaining spectral coherence during fusion, even when the attention is calculated across spatial tokens. Through this strategy, the attention block not only be more robust to misalignment and noise but correspondingly aligns sufficiently with the underlying spectral structure of the scene. Following the same approach of (15), we calculate $\mathbf{A}_{ms}^{\mathrm{cross}}$ and after then normalize the outputs as the following:

$$\begin{aligned} \mathbf{Z}_1' &= \mathrm{Norm}\left(\mathbf{A}_{hs}^{\mathrm{cross}}\right) \\ \mathbf{Z}_2' &= \mathrm{Norm}\left(\mathbf{A}_{ms}^{\mathrm{cross}}\right). \end{aligned} \tag{16}$$

The final output of the offered DDuA transformer block, Fused-F ($\mathbf{Z}_{out}$), is acquired by refining the $\mathbf{Z}_1'$ and $\mathbf{Z}_2'$ through the FFN and utilizing the skip connection to accelerate the training process. This process can be expressed as

$$\mathbf{Z}_{out} = \mathbf{W}\left(\left(\mathrm{FFN}\left(\mathbf{Z}_1'\right)+\mathbf{Z}_1'\right)+\left(\mathrm{FFN}\left(\mathbf{Z}_2'\right)+\mathbf{Z}_2'\right)\right) \tag{17}$$

where **W** is a learnable projection matrix.

D. Feature Accumulation and Reconstruction Block

As illustrated in Fig. 1, our proposed network processes the observed pair into various scales to enhance the capture of much of the intrinsic information. Specifically, the number of scales is equal to 3 in this study: the *original*, *half*, and *quarter* size. Following the outline in Fig. 1, at each feature scale, the suggested DACTM block takes the two input features and generates merged information with the same spatial size as the processing scale. In this work, three outputs, F1, F2,

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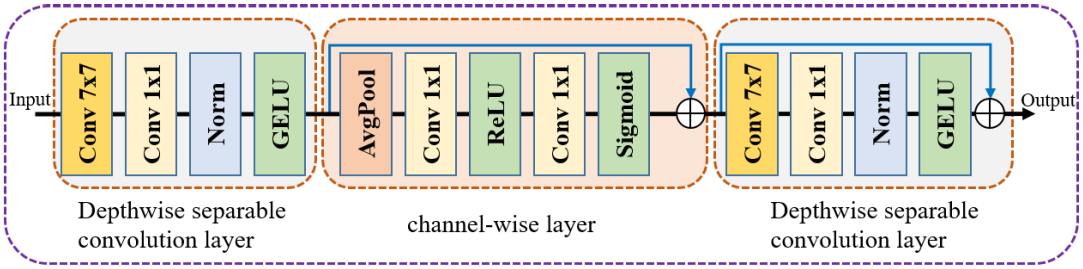


Fig. 7. Details of the FARB.

and F3, are acquired by employing the DACTM block on three different scale inputs. The final desired fused image is then reconstructed gradually, starting from the smallest scale (F3), by utilizing the proposed FARB. FARB is a straightforward but practical module that improves the fusion results besides adjusting the number of spectral bands to the same as the desired HrHSI. This block is constructed by two depthwise separable convolution layers (DWConv) and one channel-wise layer in between. Fig. 7 depicts the detailed processes of this module. Instead of directly processing the DACTM block's output, the proposed FARB processes the concatenation of each output with its corresponding input pair to benefit from the inputs' residual information. This process can be presented as

$$\text{FuScale3} = \text{FARB}\big(cat\big(\text{F3}, \text{LrScale3}, \text{HrScale3}\big)\big) \tag{18}$$

where $cat(\cdot)$ denotes concatenation operation. After that, the output of FARB at the smallest scale is upscaling (here, we used a bilinear filter) to the direct upper scale, concatenating with the direct upper scale's output and inputs to apply FARB again. We then followed this strategy until we constructed the FuScale1. Finally, the output of the proposed TM-MOE network is reconstructed as follows:

$$\widetilde{\mathbf{X}} = \text{FuScale1} + \text{up}\big(\mathbf{Y}\big) \tag{19}$$

where $\text{up}(\cdot)$ is the upscaling process utilizing a bilinear filter.

E. Loss Function

Given the observed pair ($\mathbf{Y}$ and $\mathbf{Z}$) as inputs, the loss function of the proposed network is calculated as

$$L(\Theta) = \frac{1}{T}\sum_{t}\left\| f_{\text{TM-MOE}}\left(\mathbf{Y}^{(t)}, \mathbf{Z}^{(t)}\right) - \mathbf{X}^{(t)}\right\|_1 + \frac{\lambda}{T}\sum_{t}\text{SAM};\left(f_{\text{TM-MOE}}\left(\mathbf{Y}^{(t)}, \mathbf{Z}^{(t)}\right), \mathbf{X}^{(t)}\right) \tag{20}$$

where $\mathbf{X}$ and T represent the ground truth and the number of training samples, respectively. $\|\cdot\|_1$ is l_1 norm that decreases the discrepancy between the output of the proposed TM-MOE network and the ground truth. To improve the spectral preservation of the fused images by our proposed network, we introduce the spectral angle mapper (SAM($\cdot$)) function as part of the loss function (the second term of the loss function). The contribution of the two terms of the loss function is controlled by λ (we set $\lambda = 0.3$ in this article). Since the proposed network (TM-MOE) is an end-to-end network, all the trainable parameters (Θ) are learned jointly from the training dataset.

IV. Experiments

This section provides the details of the experimentations that prove the performance of the offered TM-MOE network compared to the state-of-the-art HS-MS image fusion techniques. Particularly, the testing datasets, evaluation numerical metrics, implementation details, and qualitative and quantitative experimental outcomes are expressed in this part. Moreover, this section goes through an ablation study.

A. Experimental Datasets

The effectiveness of the suggested TM-MOE network is affirmed by utilizing three publicly available remote sensing datasets, which are widely engaged in the HS-MS image fusion task. Specifically, these datasets are: Houston,[1] Chikusei,[2] and Pavia Centre[3] datasets, which are briefly described in the following points.

1) *Houston Dataset:* This dataset was remotely sensed in February 2017 using the Optech Titam multiwave (MW) (14SEN/CON340) detector's LiDAR and ITRES CASI-1500 HS imaging instrument over the campus of the University of Houston, which was issued during the 2018 IEEE GRSS Data Fusion Challenge under the name of Houston 2018 hyperspectral dataset. Split into 50 spectral bands, the HSI of Houston 2018 covers the wavelength spectrum 380 to 1050 nm. Notably, we preserve 46 spectral channels for our experimentations and remove the four low SNR spectral channels. The spatial dimension of the Houston 2018 HSI is 601×2384 pixels. For the testing phase, 512×512 pixels are clipped without overlap with the remaining portion, which is employed for the training procedure.
2) *Chikusei Dataset:* Employing a near-infrared (NIR) and visible HS imaging apparatus, the remote sensing Chikusei HSI was obtained on 29 July 2014 from farming and metropolitan areas of Chikusei, Ibaraki, Japan. This HSI enfolds a spectral range spanning from 363 to 1018 nm, which is separated into 128 spectral channels. The Chikusei HSI has a spatial size of 2517×2335 pixels, where three 512×512 images are maintained for testing. The suggested network is trained on this data using the remaining pixels without overlaying with the test regions. It is noteworthy that the training and

[1]Houston 2018 remote sensing dataset.
[2]Chikusei remote sensing dataset.
[3]Pavia Centre remote sensing dataset.

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TABLE I
DIMENSIONS OF THE SMALL CUBES USED FOR TRAINING THE THREE TESTING DATASETS

Scale	4			8		
Data	LrHSI	HrMSI	HrHSI	LrHSI	HrMSI	HrHSI
Houston	$16 \times 16 \times 46$	$64 \times 64 \times 4$	$64 \times 64 \times 46$	$8 \times 8 \times 46$	$64 \times 64 \times 4$	$64 \times 64 \times 46$
Chikusei	$8 \times 8 \times 128$	$32 \times 32 \times 4$	$32 \times 32 \times 128$	$4 \times 4 \times 128$	$32 \times 32 \times 4$	$32 \times 32 \times 128$
Pavia	$8 \times 8 \times 93$	$32 \times 32 \times 4$	$32 \times 32 \times 93$	$4 \times 4 \times 93$	$32 \times 32 \times 4$	$32 \times 32 \times 93$

testing datasets are extracted from the halfway part, which occupies a region of 2048×2048 pixels, where the borders with low spatial resolution are scrapped.

3) *Pavia Centre Dataset:* The remote sensing hyperspectral image of Pavia Centre was captured over Pavia, northern Italy, employing the ROSIS imagery instrument during a flight campaign. This dataset holds 1096×1096 pixels with a ground resolution of 1.3 m. The spectral resolution of Pavia Centre dataset is 102 spectral channels ranging from 430 to 860 nm. After discarding the spectral bands with low SNR, 93 spectral channels survive for our experimentation. Moreover, some pixels possess no information, which are withdrawn, and the remaining pixels are utilized for the investigation; among them, 512×512 pixels are operated for testing without overlapping with training data.

B. Implementation Details

Because there are no ground-truth images with high spatial resolution for the above-mentioned remote-sensing HSIs, and the suggested TM-MOE network is an end-to-end supervised framework that requires references for training, we simulate the LrHSI and HrMSI from the original HSI operating (1) and (2), where the original HSIs will serve as ground-truth. Precisely, the LrHSIs for these three remote-sensing HSIs are simulated by employing a 7×7 Gaussian kernel (where mean = 0, and standard deviation = 2) on the original observed HSIs. Then, the spatial dimension is downsampled by factors of 4 and 8. Furthermore, we simulate the HrMSIs for Houston HSI by operating the WorldView satellite's spectral response function, while for Chikusei and Pavia Centre datasets by employing the spectral response function of the IKONOS sensor. Moreover, the simulated HrMSIs and LrHSIs are deteriorated by adding i.i.d. Gaussian noise with SNR equal to 30 to complicate the learning duty, which leads to better confirming the significance of the presented technique. We broke the training data into small cubes to increase the training pairs, as shown in Table I. The proposed network is implemented utilizing Pytorch 1.12.1 on a 64-bit Linux system environment and Python 3.12. The employed PC has a RAM of 32 GB and an Nvidia 4090 GPU. The training phase is running for 100 epochs and optimized via the Adam algorithm with a learning rate equal to $1e^{-3}$, which is halved every 30 epochs.

C. Comparison Approaches

To additionally evaluate the efficacy and significance of the offered degradation-aware hybrid transformer-Mamba network (TM-MOE) for HS and MS image fusion, a vast comparison analysis and discussion are provided by comparing it with seven state-of-the-art HS and MS image fusion approaches, including: nonnegative-structured sparse representation (NSSR) technique [16],[4] the deep hyperspectral image-sharpening (DHSIS) technique [37],[5] coupled NMF (CNMF) technique [10],[6] deep spatiospectral attention CNN (HSRNet) technique [25],[7]dual spatial–spectral pyramid network with transformer (DSPNet) technique [44],[8] a novel pyramid shuffle-and-reshuffle transformer (PSRT) technique [23],[9] and cyclic cross-modality interaction (CYformer) technique [12].[10] Notably, CNMF and NSSR are model-based approaches with no need for training, while the rest of the comparison techniques are learning-based fusion frameworks. Herein, the learning-based strategies are retrained following the optimal hyperparameters suggested by their original articles. The testing and training data are processed identically as used in the proposed method for fair comparison.

D. Quantitative Evaluation Indices

Five widely utilized quantitative metrics are nominated to assess the performance of the suggested TM-MOE for HS and MS image fusion against the comparison approaches. Particularly, we pick the relative dimension global error in synthesis (ERGAS), peak signal-to-noise ratio (PSNR), universal image quality index (UIQI), structural similarity index (SSIM), and spectral angle mapper (SAM). Given the fused image $\widetilde{\mathbf{X}}$ and its ground-truth $\mathbf{X}$, ERGAS estimates the overall error across the entire spectral channels of the acquired fused image compared to its reference, which mathematically can be expressed as follows:

$$ERGAS\left(\mathbf{X},\widetilde{\mathbf{X}}\right) = \frac{100}{d}\sqrt{\frac{1}{L}\sum_{i=1}^{L}\frac{mse\left(\mathbf{X}^{(i)},\widetilde{\mathbf{X}}^{(i)}\right)}{\langle\widetilde{\mathbf{X}}^{(i)}\rangle^{2}}} \tag{21}$$

where mse is mean squared error, $\langle\cdot\rangle$ represents the mean calculated over all pixels of $\widetilde{\mathbf{X}}^{(i)}$, and d denotes the spatial downsampling factor. *zero* is the best value of ERGAS. PSNR assesses the spectral distortion of predicted HrHSI, where the higher its value, the better, which can be expressed as

$$PSNR\left(\mathbf{X},\widetilde{\mathbf{X}}\right) = \frac{1}{L}\sum_{i=1}^{L}PSNR\left(\mathbf{X}^{(i)},\widetilde{\mathbf{X}}^{(i)}\right) \tag{22}$$

[4]https://see.xidian.edu.cn/faculty/wsdong/HSI_SR_project.htm
[5]https://github.com/renweidian/DHSIS
[6]http://naotoyokoya.com/Download.html
[7]https://liangjiandeng.github.io/Projects_Res/HSRnet_2021tnnls.html
[8]https://github.com/syc11-25/DSPNet
[9]https://github.com/Deng-shangqi/PSRT
[10]https://github.com/Tomchenshi/CYformer

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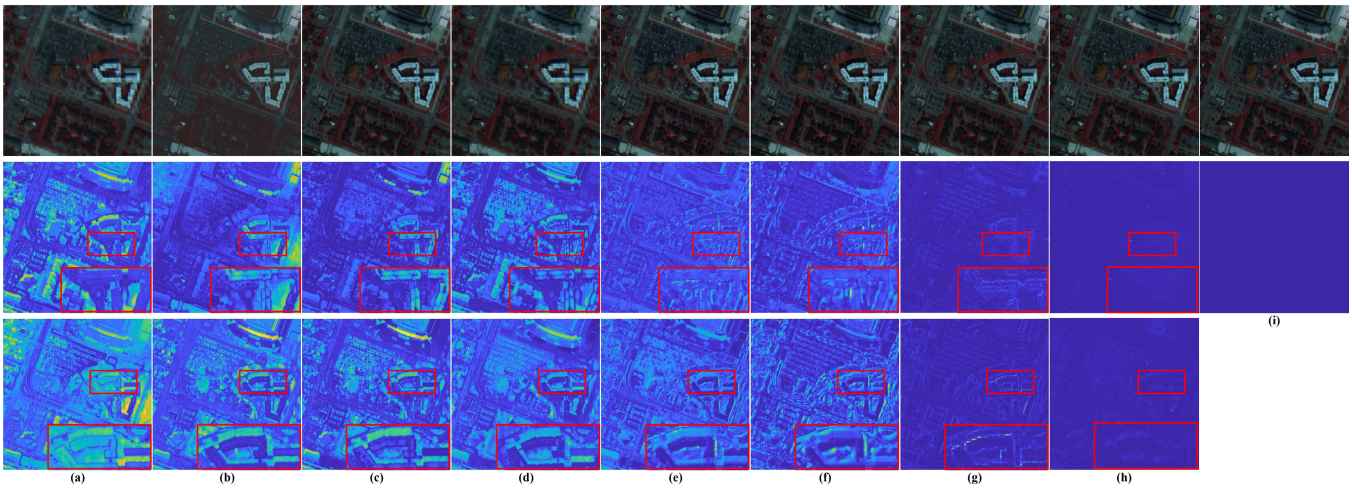


Fig. 8. Visual outcomes (false color composite R = 26, G = 15, and B = 10) from the Houston dataset (scale factor 8) acquired by: (a) CNMF, (b) NSSR, (c) DHSIS, (d) HSRNet, (e) DSPNet, (f) PSRT, (g) CYformer, (h) TM-MOE, and (i) ground-truth. The second and third rows display the mean squared error and spectral angle mapper heat maps, respectively, between the achieved results and the reference image.

TABLE II

NUMERICAL EVALUATION ATTAINMENTS [PSNR (DB), SAM, ERGAS, UIQI, AND SSIM] OF EXPERIMENTATIONS PERFORMED ON THE HOUSTON DATASET FOR THE SUGGESTED TM-MOE NETWORK FOR HS AND MS IMAGE FUSION AND ITS COMPETITORS OF STATE-OF-THE-ART FUSION STRATEGIES (CNMF, NSSR, DHSIS, HSRNET, DSPNET, PSRT, CYFORMER). THE MOST SATISFACTORY UPSHOT IS NOTED IN BOLD

Method (s)	4×					8×				
	PSNR (dB)	SAM	ERGAS	UIQI	SSIM	PSNR (dB)	SAM	ERGAS	UIQI	SSIM
Ideal value	∞	0	0	1	1	∞	0	0	1	1
CNMF [10]	36.1840	5.0460	4.9760	0.9265	0.9537	33.8943	6.2555	6.5570	0.9082	0.9486
NSSR [16]	38.1418	3.9881	4.6501	0.9395	0.9626	36.0718	4.5695	5.2229	0.9228	0.9560
DHSIS [37]	39.7798	2.9980	3.2953	0.9412	0.9605	36.7519	3.9682	3.5204	0.9355	0.9585
HSRNet [25]	41.1453	2.9609	2.8034	0.9549	0.9748	38.5860	3.1927	3.0962	0.9482	0.9691
DSPNet [44]	45.8122	2.0139	1.4096	0.9830	0.9857	42.8126	2.4930	1.6152	0.9796	0.9829
PSRT [23]	45.3199	2.3786	1.5734	0.9809	0.9862	40.6309	2.8381	1.8819	0.9782	0.9813
CYformer [12]	46.9726	1.7354	1.1108	0.9857	0.9891	44.3857	1.9481	1.4937	0.9834	0.9864
TM-MOE	**49.0759**	**1.1835**	**0.6106**	**0.9926**	**0.9940**	**46.7253**	**1.4759**	**0.8175**	**0.9901**	**0.9923**

where ∞ is the best amount of PSNR with no spectral deterioration. On the other hand, UIQI is estimated over the sliding window, after which it computes the mean across the entire spectral bands and windows as follows:

$$UIQI\left(\mathbf{X},\widetilde{\mathbf{X}}\right)=\frac{4\mu_{\mathbf{X}}\mu_{\widetilde{\mathbf{X}}}}{\mu_{\mathbf{X}}^2+\mu_{\widetilde{\mathbf{X}}}^2}\frac{\sigma_{\mathbf{X},\widetilde{\mathbf{X}}}^2}{\sigma_{\mathbf{X}}^2+\sigma_{\widetilde{\mathbf{X}}}^2} \tag{23}$$

herein, σ and μ denote variance and mean, respectively. UIQI ranges from 0 to 1, where 1 is the finest amount. Moreover, we utilize the SSIM index to assess the spatial distortion, where its mean over the entire channels is calculated as

$$SSIM\left(\mathbf{X},\widetilde{\mathbf{X}}\right)=\frac{1}{L}\sum_{i=1}^{L}SSIM\left(\mathbf{X}^{(i)},\widetilde{\mathbf{X}}^{(i)}\right) \tag{24}$$

and the adequate worth of SSIM is 1, which implies no spatial deterioration. The last quantitative index used in this article is SAM, which assesses the absolute amount of the spectrum gradients (degrees) between the constructed fused image and its ground truth as follows:

$$SAM\left(\mathbf{X},\widetilde{\mathbf{X}}\right)=\frac{1}{L}\sum_{i=1}^{L}arcos\frac{\mathbf{X}^{(i)^{\mathrm{T}}}\widetilde{\mathbf{X}}^{(i)}}{\|\mathbf{X}^{(i)}\|_2\|\widetilde{\mathbf{X}}^{(i)}\|_2} \tag{25}$$

where T is the transpose operation. The best indicator for SAM is 0.

E. Experimental Results

1) *Empirical Outcomes on the Houston HSIs*: The upshots of the diverse numerical investigations performed on the Houston dataset are depicted in Table II. For clarity, the most relevant upshots are emphasized in bold. Table II plainly reveals the adequate performance of the proposed TM-MOE approach that outperforms various state-of-the-art fusion techniques. All the learning frameworks gain more satisfactory consequences compared to classical approaches, with varying performance among them due to the strength of characteristic extraction. For example, the suggested TM-MOE network attains the most significant intensity distribution with respect to the competing techniques, according to its best achievement of SAM. This is due to the qualification of the suggested technique in terms of dragging the intrinsic characteristics and rebuilding the fused information. Moreover, we pictured the fused HrHSIs attained from various comparative strategies, including the offered TM-MOE, in Fig. 8. According to Fig. 8, the CNMF, DHSIS, and HSRNet suffer from texture antiques. In the same formulation, NSSR has spectral distortion and blurring, while DSPNet and PSRT are relatively bright. To further highlight attainment, the second and third rows of Fig. 8

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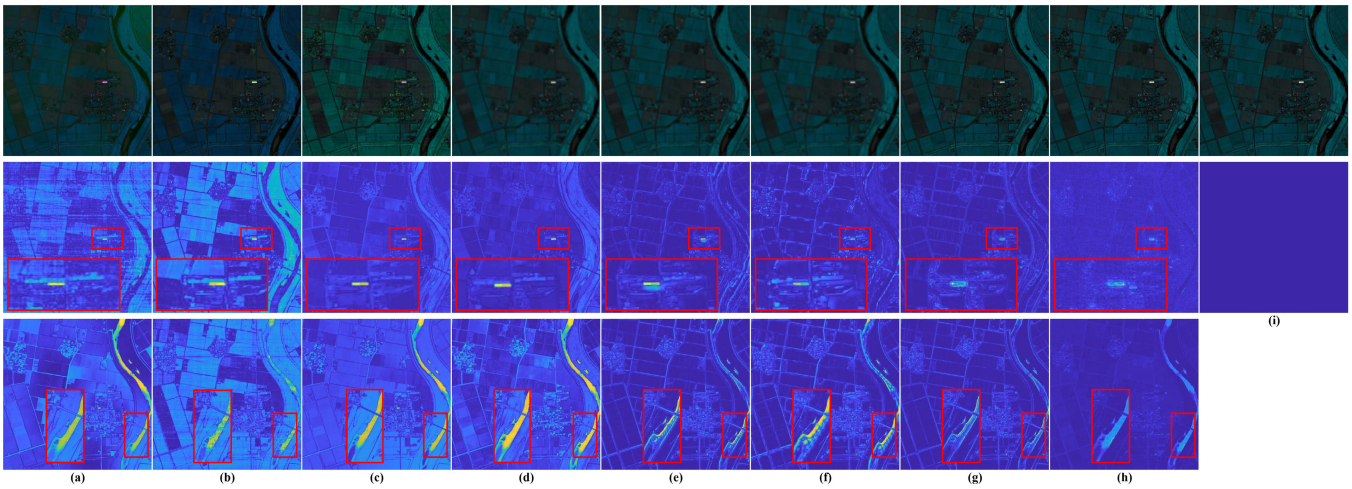


Fig. 9. Visual outcomes (false color composite R = 60, G = 75, and B = 90) from the Chikusei dataset (scale factor 4) acquired by: (a) CNMF, (b) NSSR, (c) DHSIS, (d) HSRNet, (e) DSPNet, (f) PSRT, (g) CYformer, (h) TM-MOE, and (i) ground-truth. The second and third rows display the mean squared error and spectral angle mapper heat maps, respectively, between the achieved results and the reference image.

portray the heat maps of the root mean square error (RMSE) and SAM between the fused results of all testing approaches and the ground-truth, which obviously reveal the efficacy of the suggested technique and its capability to rebuild and attain a fused image closest to the reference image against the testing scenarios.

2) *Empirical Outcomes on the Chikusei HSIs*: Table III displays the quantitative indicators, particularly PSNR, SAM, ERGAS, UIQI, and SSIM, of the evaluation approaches encompassing the TM-MOE network on the Chikusei data, where the optimal acquired outcomes are written in bold. As depicted in Table III, the suggested TM-MOE fusion strategy earns the most subordinate dynamic shift and modification between its fused upshot and the reference image, in comparison with the comparative approaches. This is indicated by the least acquired value of ERGAS of the network proposed in this study. Moreover, our proposed approach achieves preferred outcomes for the other indices, resulting in better structure preservation and minimal spectral distortion when compared to other competing techniques. Furthermore, the analysis of performance is more thoroughly investigated by displaying the visual outcomes of the testing techniques, as illustrated in Fig. 9. The first row of Fig. 9 depicts the false RGB color of attained fusion upshots of the evaluation approaches, including our proposed technique. Moreover, the RMSE and SAM heat maps computed from the fused image with respect to the ground truth image are portrayed in the second and third rows of Fig. 9, respectively. This figure clearly demonstrates that the proposed TM-MOE approach performs most satisfactorily in terms of image reconstruction, preserving most pleasing structural and spectral attributes compared to all competitors.

3) *Empirical Outcomes on the Pavia Centre HSIs*: To further substantiate the usefulness of the presented technique, we execute extra investigations through experiments on a third simulated dataset, the Pavia Centre HSI. The quantitative findings of the suggested TM-MOE network and the other testing frameworks are presented in Table IV, where the optimum outputs are typed in bold. As demonstrated in Table IV, the strategy presented in this study yields optimal outcomes in comparison with the competing approaches. For example, the lowest quantity of SAM and ERGAS, and the most significant values of PSNR, UIQI, and SSIM are gained by the suggested TM-MOE network. These better outcomes result in enhanced spatial structure maintenance, accompanied by the lowest dynamic shift and change, as indicated by the highest values of SSIM and the minimum quantity of ERGAS for the proposed TM-MOE, respectively. Furthermore, the graphical achievements are depicted in Fig. 10, which comprises false RGB color for the acquired fused images of the various testing strategies on the first row, the residual between the fused images and the reference image on the second row, and on the third row, the heat maps of SAM calculated between the obtained fused image and the ground-truth image. The visual outcomes, as displayed in Fig. 10, are consistent with the numerical outputs, which prove the capability of the suggested TM-MOE in terms of lower spectral distortion, better texture detail preservation, and the closest visual appearance result to the ground truth among all competing and testing techniques.

To promote performance investigation, we calculate the PSNR and SSIM per band for the fused images attained by the competing practices. The acquired outcomes are displayed in Figs. 11 and 12. As portrayed in Fig. 11, the PSNR per band of the devised TM-MOE network conveys the highest PSNR value for most of the bands of the three empirical datasets. Furthermore, there are a few PSNR scores of the TM-MOE network that do not earn the optimal value across the competing approaches, but still demonstrate exemplary performance. Fig. 11 obviously reveals the capability of the proposed approach in spectral information maintenance.

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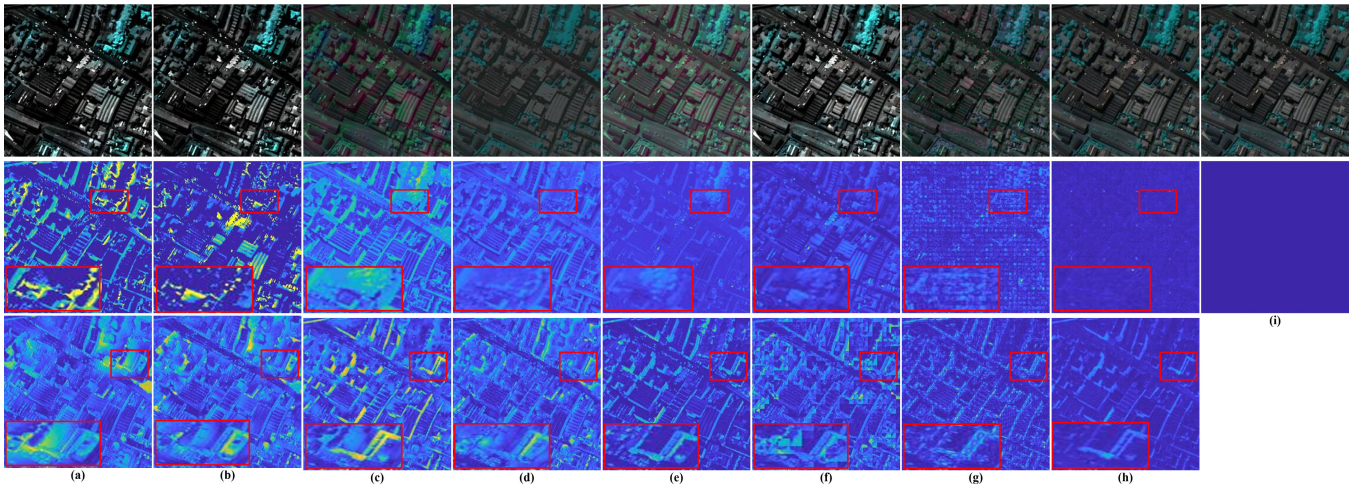


Fig. 10. Visual outcomes (false color composite R = 60, G = 75, and B = 90) from the Pavia Centre dataset (scale factor 8) acquired by: (a) CNMF, (b) NSSR, (c) DHSIS, (d) HSRNet, (e) DSPNet, (f) PSRT, (g) CYformer, (h) TM-MOE, and (i) ground-truth. The second and third rows display the mean squared error and spectral angle mapper heat maps, respectively, between the achieved results and the reference image.

TABLE III

AVERAGE NUMERICAL EVALUATION ATTAINMENTS [PSNR (DB), SAM, ERGAS, UIQI, AND SSIM] OF EXPERIMENTATIONS PERFORMED ON THE CHIKUSEI DATASET FOR THE SUGGESTED TM-MOE NETWORK FOR HS AND MS IMAGE FUSION AND ITS COMPETITORS OF STATE-OF-THE-ART FUSION STRATEGIES (CNMF, NSSR, DHSIS, HSRNET, DSPNET, PSRT, AND CYFORMER). THE MOST SATISFACTORY UPSHOT IS NOTED IN BOLD

Method (s)	4×					8×				
	PSNR (dB)	SAM	ERGAS	UIQI	SSIM	PSNR (dB)	SAM	ERGAS	UIQI	SSIM
Ideal value	∞	0	0	1	1	∞	0	0	1	1
CNMF [10]	35.8859	5.3762	6.5289	0.8964	0.9427	33.4117	6.4244	7.8911	0.8648	0.9338
NSSR [16]	34.3460	5.6341	6.1785	0.8709	0.9371	31.7872	6.9783	7.4963	0.8520	0.9261
DHSIS [37]	37.2673	4.3135	4.9043	0.9297	0.9514	35.7636	5.1182	5.3520	0.9164	0.9475
HSRNet [25]	39.6418	3.5628	3.9452	0.9419	0.9578	37.8343	3.8613	4.4906	0.9359	0.9524
DSPNet [44]	44.2390	2.5125	2.7747	0.9742	0.9820	43.1074	2.8996	2.9485	0.9701	0.9795
PSRT [23]	43.8527	2.6932	2.6495	0.9713	0.9785	41.5449	3.2658	2.9719	0.9652	0.9736
CYformer [12]	45.4156	2.3781	2.4339	0.9778	0.9867	44.1631	2.5804	2.7184	0.9746	0.9842
TM-MOE	**47.6974**	**1.9778**	**2.0955**	**0.9822**	**0.9918**	**46.9413**	**2.2221**	**2.3643**	**0.9802**	**0.9895**

TABLE IV

NUMERICAL EVALUATION ATTAINMENTS [PSNR (DB), SAM, ERGAS, UIQI, AND SSIM] OF EXPERIMENTATIONS PERFORMED ON THE PAVIA CENTRE DATASET FOR THE SUGGESTED TM-MOE NETWORK FOR HS AND MS IMAGE FUSION AND ITS COMPETITORS OF STATE-OF-THE-ART FUSION STRATEGIES (CNMF, NSSR, DHSIS, HSRNET, DSPNET, PSRT, AND CYFORMER). THE MOST SATISFACTORY UPSHOT IS NOTED IN BOLD

Method (s)	4×					8×				
	PSNR (dB)	SAM	ERGAS	UIQI	SSIM	PSNR (dB)	SAM	ERGAS	UIQI	SSIM
Ideal value	∞	0	0	1	1	∞	0	0	1	1
CNMF [10]	32.2708	7.4301	7.2625	0.8624	0.9475	30.7169	8.1382	7.8359	0.8461	0.9381
NSSR [16]	33.9742	6.5268	5.9659	0.8726	0.9519	32.2935	7.1690	6.5826	0.8614	0.9425
DHSIS [37]	36.1963	4.7786	4.4528	0.9361	0.9574	34.9370	5.3584	4.9479	0.9246	0.9493
HSRNet [25]	38.0895	4.2904	4.2419	0.9487	0.9613	36.5193	4.6997	4.8457	0.9353	0.9558
DSPNet [44]	41.8754	3.1428	2.4673	0.9802	0.9816	40.4598	3.6433	2.8269	0.9790	0.9790
PSRT [23]	41.6831	3.2199	2.5330	0.9796	0.9821	40.3852	3.8176	2.7884	0.9772	0.9794
CYformer [12]	43.2907	2.9643	1.8456	0.9865	0.9880	41.7196	3.2779	1.9158	0.9827	0.9837
TM-MOE	**44.0183**	**2.5145**	**1.1271**	**0.9907**	**0.9935**	**42.9758**	**2.8792**	**1.2841**	**0.9894**	**0.9902**

Moreover, in terms of the spatial details preservation, the SSIM per bands offered in Fig. 12, which exhibits the capacity of the suggested TM-MOE strategy in this regard, due to the best SSIM scores acquired for most of the bands of the three testing datasets. In summary, the presented TM-MOE exhibits its effectiveness in preserving both spatial and spectral characteristics due to its capability of conveying the intrinsic information through the collaborative interaction between its proposed components. The proposed TM-MOE benefits from the transformer's ability to better exploit nonlocal features. Moreover, this transformer is further improved by incorporating a spectral affinity prior to reduce spectral distortion. On the other hand, the devised technique incorporates the 3-D Mamba block to improve feature extraction and also leverages the efficiency of the state-space model for modeling relationships. Through this combination, the proposed fusion technique efficiently outperforms state-of-the-art methods, as shown in the experimental results. Furthermore, as shown

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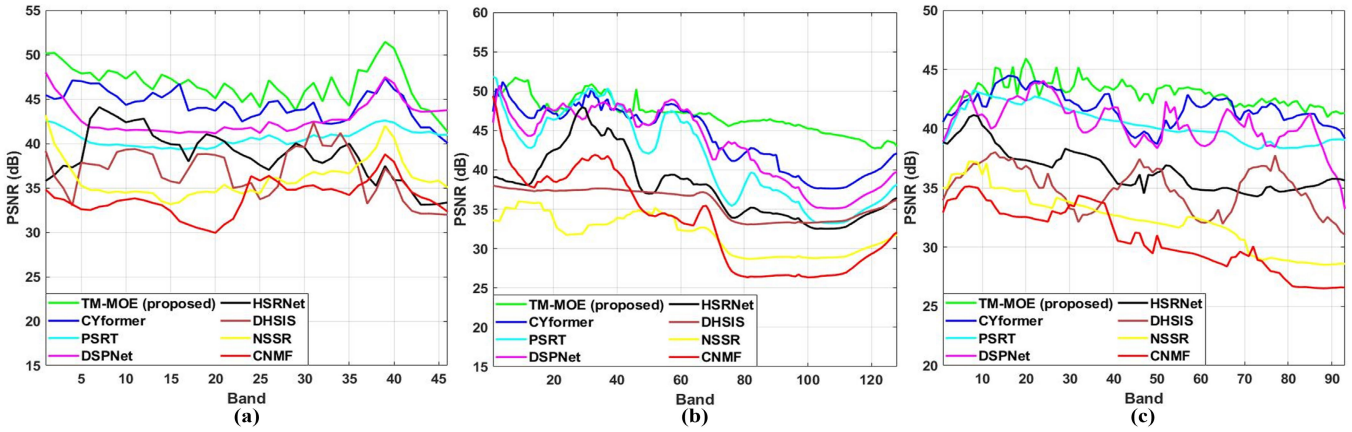


Fig. 11. Per spectral channel efficacy of the testing strategies in terms of PSNR of the experimentations on: (a) Houston dataset, (b) Chikusei dataset, and (c) Pavia Centre dataset.

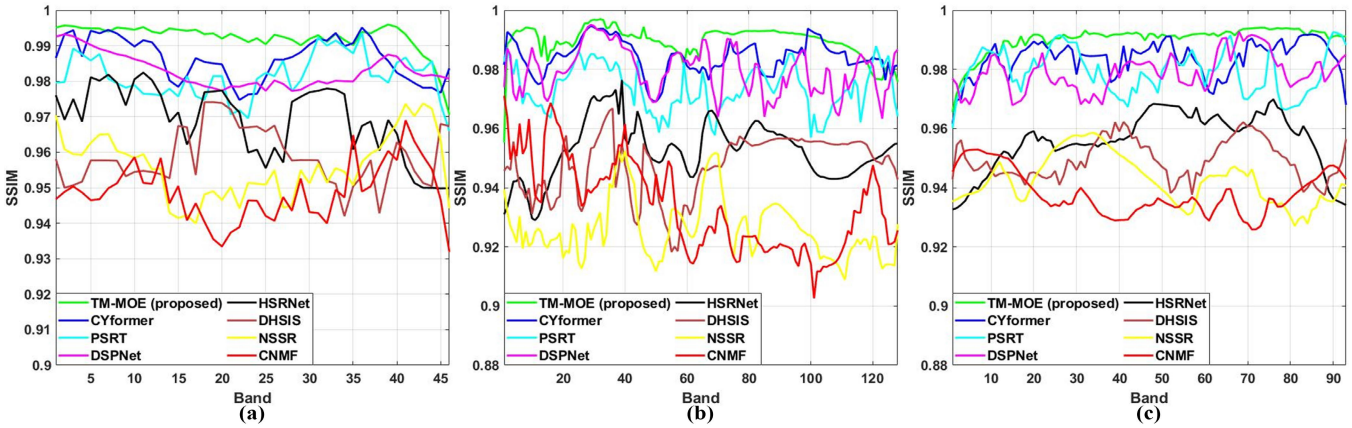


Fig. 12. Per spectral channel efficacy of the testing strategies in terms of SSIM of the experimentations on: (a) Houston dataset, (b) Chikusei dataset, and (c) Pavia Centre dataset.

in Section IV-F, the devised TM-MOE technique effectively balances between complexity and performance.

F. Computational Comparison of the Various Testing Techniques

In this section, we examine the computational complexity of the various approaches by providing their number of trainable parameters and FLOPs alongside their attainment in terms of PSNR. Notably, the classical testing (CNMF and NSSR) approaches are excluded in this investigation due to the nature of this class's strategies, which require more computation and possess manual parameters that are challenging to tune. Although DHSIS has a step that utilizes a CNN, it employs two analytical solutions: one pre- and one postlearning of image priors through the CNN model, which makes it more complex and still requires hand-crafted parameters. However, the complexity of the end-to-end competing approaches, including the proposed TM-MOE network, is depicted in Fig. 13. As displayed in Fig. 13, PSRT and HSRNet have the lowest number of parameters and FLOPs among all learning-based techniques, yet they have the worst performance across all DL-based methods. DSPNet achieves better outcomes than PSRT and HSRNet, with a significant increase in both the number of parameters and FLOPs compared to all DL-based testing methods. Although the competing model CYformer has a lower number of parameters compared to the suggested TM-MOE, TM-MOE achieves fewer FLOPs with significant results improvement. In summary, Fig. 13 clearly describes the efficiency and effectiveness of the devised TM-MOE. Furthermore, we empirically stated the order of magnitude of forward inference time and peak memory consumption of the devised method and the tested DL approaches (HSRNet, DSPNet, PSRT, and CYformer) under the same hardware and implementation settings. In this regard, PyTorch 1.12.1 on a 64-bit Linux environment and Python 3.12 are used for implementation on a PC with 32 GB of RAM and an Nvidia 4090 GPU. We use testing images from the Houston dataset with dimensions of ($2 \times 46 \times 64 \times 64$ and $2 \times 4 \times 256 \times 256$) as inputs, and output ($2 \times 46 \times 256 \times 256$). As shown in Table V, PSRT has the lowest inference time, but its performance is worse than that of CYformer and the devised method in this article. On the other hand, HSRNet achieved the optimal value of memory usage. However, its results across all testing datasets are worse than those of all DL techniques, including the proposed technique. Briefly, across the testing datasets,

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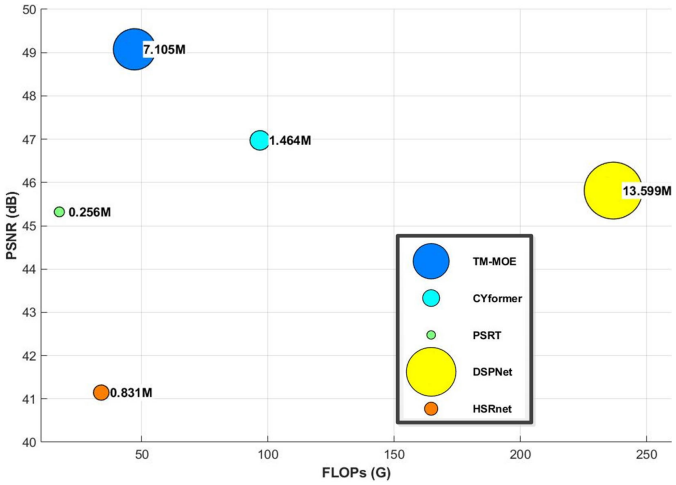


Fig. 13. Complexness of various competing approaches in terms of the number of parameters and FLOPs compared to their performance.

TABLE V

ORDER-OF-MAGNITUDE FORWARD INFERENCE TIME (MS), PEAK MEMORY (MB) CONSUMPTION, AND PSNR (DB) OF THE TM-MOE NETWORK AND COMPETING DL FRAMEWORKS (HSRNET, DSPNET, PSRT, AND CYFORMER) FOR HS-MS IMAGE FUSION. THE OPTIMAL VALUES IS NOTED IN BOLD

Method	HSRNet	DSPNet	PSRT	CYformer	TM-MOE
Time	276.31	1445.53	**248.21**	783.40	506.45
Memory	**187.73**	1294.05	701.15	2521.69	1561.79
PSNR (dB)	41.1453	45.8122	45.3199	46.9726	**49.0759**

* Higher PSNR indicates better reconstruction quality. Scale factor = $\times 4$.

TABLE VI

NUMERICAL EVALUATIONS OF THE INFLUENCE OF ASSORTED PROPOSED MODULES ON THE EFFECTIVENESS OF THE SUGGESTED TM-MOE NETWORK FOR HS-MS IMAGE FUSION, THE OPTIMAL OUTCOMES ARE NOTED IN BOLD

Method	$4\times$				
	PSNR (dB)	SAM	ERGAS	SSIM	Params (M)
Best value	∞	0	0	1	–
Net1	47.6817	1.5131	0.7045	0.9912	7.083
Net2	44.8793	1.7613	0.8529	0.9875	**2.760**
Net3	45.2674	1.8409	0.8497	0.9882	6.202
TM-MOE	**49.0759**	**1.1835**	**0.6106**	**0.9940**	7.105

CYformer is most competitive with the proposed technique (TM-MOE), while TM-MOE achieves the best results, lower inference time, and lower peak memory consumption. By this means, the proposed TM-MOE technique can effectively balance the complexity and performance.

G. Ablation Study

In this section, we investigate the contribution and capability of each principal component of the devised TM-MOE in terms of its effectiveness. Our developed fusion technique comprises three 3-D scan Mamba modules, a constrained cross-attention transformer, and a reconstructor unit, as displayed in Fig. 1. Thus, three distinct architectures are tested: the suggested TM-MOE without the Mamba blocks (Net1), TM-MOE without the transformer (Net2), and TM-MOE without the reconstructor block (Net3). Net1 bans the two Mamba blocks placed before the transformer for characteristic extraction, while all further elements stay untouched. Here, we involved two convolutional layers for feature extraction instead. In Net2, the transformer block is removed and substituted by a convolution layer, with the concatenation of the observed pair features returning one image. Net3, instead of the reconstructor block, employs a convolutional layer to combine the fused images at various scales and adapt the number of desired spectral bands. The upshots of the investigations of the various architectures are represented in Table VI. As illustrated in Table VI, each of the devised parts of our TM-MOE technique contributes enormously to its performance. Notably, the 3-D scan Mamba block has the lowest number of parameters, which enhances all results due to its capability of characteristic extraction in an efficient manner. On the other hand, although it has a higher number of parameters compared to the other suggested components, the constrained cross-attention transformer improves the performance immensely, which is attributed to its capacity to fuse the latent information from both input images, HrMSI and LrHSI. Furthermore, the upshots attained with the absence of the reconstructor block (FARB) indeed indicate its importance for rebuilding the finest details of HrHSI. Briefly, the suggested TM-MOE that incorporates these devised modules gains good outcomes with a manageable number of parameters.

TABLE VII

NUMERICAL EVALUATIONS OF THE INFLUENCE OF THE MIXTURE-OF-EXPERTS (MOE) NETWORK ON THE EFFECTIVENESS OF THE SUGGESTED TM-MOE NETWORK FOR HS-MS IMAGE FUSION, THE OPTIMAL OUTCOMES ARE NOTED IN BOLD

Method	$4\times$				
	PSNR (dB)	SAM	ERGAS	SSIM	Params (M)
Best value	∞	0	0	1	–
w/o MoE	48.1278	1.3995	0.6841	0.9924	**7.103**
TM-MOE	**49.0759**	**1.1835**	**0.6106**	**0.9940**	7.105

Moreover, as portrayed in Fig. 2, each Mamba block of the three suggested Mamba blocks has a layer that employs a MoE network instead of an MLP, which aids in expanding the capability of the presented framework while maintaining the active learnable parameters miniature, which leads to an enhanced computation process. To further examine the influence of the devised MoE on the performance of the suggested TM-MOE, we substitute all MoE with an MLP network (called w/o MoE) and compare its outcomes with the original TM-MOE architecture. The empirical upshots of involving MoE are presented in Table VII. Table VII clearly reveals the effectiveness of applying the MoE network, as it expands the architecture and features through a feasible computational process, leading to favorable outcomes.

V. CONCLUSION

As the HS-MS image fusion task evolves as a highly demanded way to overcome the restrictions of standard imaging sensors regarding acquiring HrHSI, this article proposes a degradation-aware transformer-Mamba MoE for HS-MS image fusion. The suggested techniques not only benefit from the advantages of the Mamba model family (e.g., efficient computation and increased throughput) and the transformer models (e.g., good performance), but also overcome their

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limitations. The core part of the suggested TM-MOE is DACTM, which exploits and fuses the details of the observed pair at various hierarchies. The DACTM block comprises two 3DMamba blocks for dragging the characteristics of the LrHSI and HrMSI, a DDuA transformer that applies cross-attention to utilize and fuse the features of both images, restraining the fusion process toward the properties of the imaging instrument, and further 3DMamba blocks to refine the fused features. While the fusion process is accomplished at various dimensions, we propose the FARB that takes the inputs and fused information at a particular scale to refine them and construct the final desired HrHSI gradually, starting from the smallest size. Diverse and wealthy experiments are performed, which obviously reveal the effectiveness and efficiency of the proposed TM-MOE technique. Future work can be divided into two aspects: improving the architecture of each block or leveraging the advantages of both models and interpreting them by developing a unified mechanism based on their concepts.

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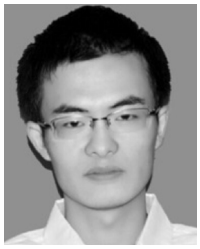
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